

Mid-Semester Project Report

1 Literature Review

The article I chose to review is *HaGRID — HAnd Gesture Recognition Image Dataset* by Kapitanov et al. (2024), the dataset and methodology on which this project is based. The paper introduces HaGRID, a large-scale hand-gesture image dataset designed for training gesture classification and dynamic gesture recognition systems. According to the abstract (page 1), the authors frame HaGRID as a resource for enabling robust gesture recognition on consumer devices, emphasizing the importance of a diverse and sufficiently large dataset for real-world performance. The dataset ultimately contains **approximately 548,000 images** covering **18 gesture classes**, including static gestures (e.g., "stop," "peace," "like") and dynamic gestures represented as separate "start/end" frames (e.g., "swipe" sequences).

The paper's main contribution is the **dataset creation pipeline**, shown in Figures 2–3 (pages 4–5). The pipeline involves (1) large-scale data collection through crowdsourcing, (2) multi-stage filtering by annotators, (3) bounding-box aggregation with consistency checks, and (4) final class annotation. The authors emphasize significant variability across subjects (age, clothing, pose), lighting conditions, backgrounds, and camera devices to improve generalization. Distribution plots on page 5 show variations in brightness, subject-to-camera distance, and resolution, supporting the authors' claim that the dataset is intentionally heterogeneous to reflect real-world scenarios.

The paper also provides extensive **baseline results** (page 6), evaluating multiple CNN-based models (e.g., ResNet-18/34/50/152, MobileNetV3, EfficientNet) on the HaGRID classification task. Reported F1-scores for static gesture classification range from ~68% to ~91%, with larger backbones generally performing better. These results demonstrate that the task is nontrivial, especially under heterogeneous lighting and subject variation. The authors further analyze the influence of dataset characteristics—lighting diversity, background uniformity, and sample density—on performance (pages 7–8), concluding that diversity is essential for robust generalization.

This paper is directly relevant to the present project for three major reasons.

1. **Dataset Alignment:** The project uses HaGRID for both RGB and landmark-based gesture recognition; therefore, understanding its collection and

annotation procedures is necessary to contextualize model performance.

2. **Baseline Comparisons:** The dataset paper's CNN baselines provide a natural point of comparison for evaluating this project's baseline CNN.
3. **Modality Relevance:** Although HaGRID itself contains images only, its high-quality bounding boxes facilitate landmark extraction (e.g., via MediaPipe Hands), enabling multimodal experimentation such as the MLP landmark model explored in this project.

Overall, the HaGRID paper offers valuable insight into dataset quality, diversity, and baseline expectations, all of which inform the challenges and outcomes of this project.

2 Baseline Experiment Design

2.1 Project Overview

The goal of this project is **hand gesture classification** using two complementary modalities:

1. **RGB images**, cropped to the hand region.
2. **Hand landmarks** (21 keypoints), normalized with respect to bounding-box coordinates.

The motivation for exploring both modalities stems from their complementary characteristics: RGB encodes texture and appearance, while landmarks encode pose and geometry.

2.2 Dataset and Preprocessing

All experiments use the **HaGRID dataset**, which includes 18 gesture classes such as "call," "fist," "palm," "peace," "stop," and their inverted or alternative variants.

Landmarks Dataset

A custom dataloader processes MediaPipe-extracted 2D landmarks.

- Training split size: **408,237 samples**.
- Landmark shape: **(21, 2)** points.
- Normalization: Landmarks are normalized with respect to the gesture bounding box, ensuring scale-invariance.

- Empty or missing-landmark samples ($\approx 5,192$) are filtered out.

RGB Dataset

The RGB dataloader loads cropped hand-region images.

- Training split size: **413,429 samples**.
- Typical image shape: **(3, 141, 128)** before resizing to model input resolution.
- Pixel values normalized to [0, 1].

The cropping step matches the dataset's own bounding-box-based structure, consistent with HaGRID's annotation pipeline shown on pages 4–5.

2.3 Baseline Models

MLP for Landmark Data

- **Input dim:** 42 (flattened 21×2 landmarks)
- **Hidden layers:** 256 $\rightarrow$ 128
- **Activation:** ReLU
- **Dropout:** 0.1
- **BatchNorm:** enabled
- **Training:**
 - learning rate = $1e-3$
 - batch size = 128
 - epochs = 8
- **Best validation accuracy: 0.9923**

This model acts as a classical baseline and aligns with common landmark-only gesture models.

CNN for RGB Data

- **Input channels:** 3
- **Conv channels:** [32, 64]
- **Kernel size:** 3
- **Pooling frequency:** every 2 conv blocks

- **Activation:** ReLU
- **Dropout:** 0.1
- **Image size:** resized to 64×64
- **Training:**
 - learning rate = 1e-3
 - batch size = 64
 - epochs = 8
- **Best validation accuracy: 0.5119**

This CNN baseline is intentionally simple and serves as a performance lower bound against pretrained architectures such as those evaluated in the HaGRID paper.

2.4 Evaluation Metrics

All experiments use:

- **Top-1 Accuracy**
- **Macro-F1 Score**
- **Macro Precision**
- **Macro Recall**
- **Full per-class classification reports**

Macro metrics are appropriate because the 18 gesture classes may exhibit slight imbalance, and macro averaging ensures each class contributes equally.

3 Preliminary Results

3.1 Result Summary

Model	Top-1 Acc	Macro-F1	Macro Prec	Macro Rec
MLP (landmarks)	0.9916	0.9916	0.9917	0.9916
CNN (RGB)	0.5198	0.5124	0.5219	0.5198

3.2 MLP Results

The MLP classifier achieves **99.16% accuracy**, with all gesture classes scoring near-perfect precision and recall. This suggests that:

- The landmark modality is extremely informative for static gestures.
- Landmark normalization successfully removes scale/position variation.
- Landmark extraction is consistent across most samples.

The only slight degradation appears in a few classes (e.g., class 5 and class 7), with F1 dropping to ~97%, but these differences are minimal.

3.3 CNN Results

The CNN baseline performs significantly worse, with accuracy around **51%**, and many gesture classes showing F1-scores in the 0.25–0.45 range. This aligns with expectations given:

1. The model is extremely shallow compared to modern CNNs.
2. Input images are downsampled to 64×64, losing fine gesture details.
3. The HaGRID paper shows (page 6) that even well-established CNNs require deep architectures and careful fine-tuning to get high performance.

3.4 Analysis

These results reveal two important findings:

(1) Landmarks are highly discriminative.

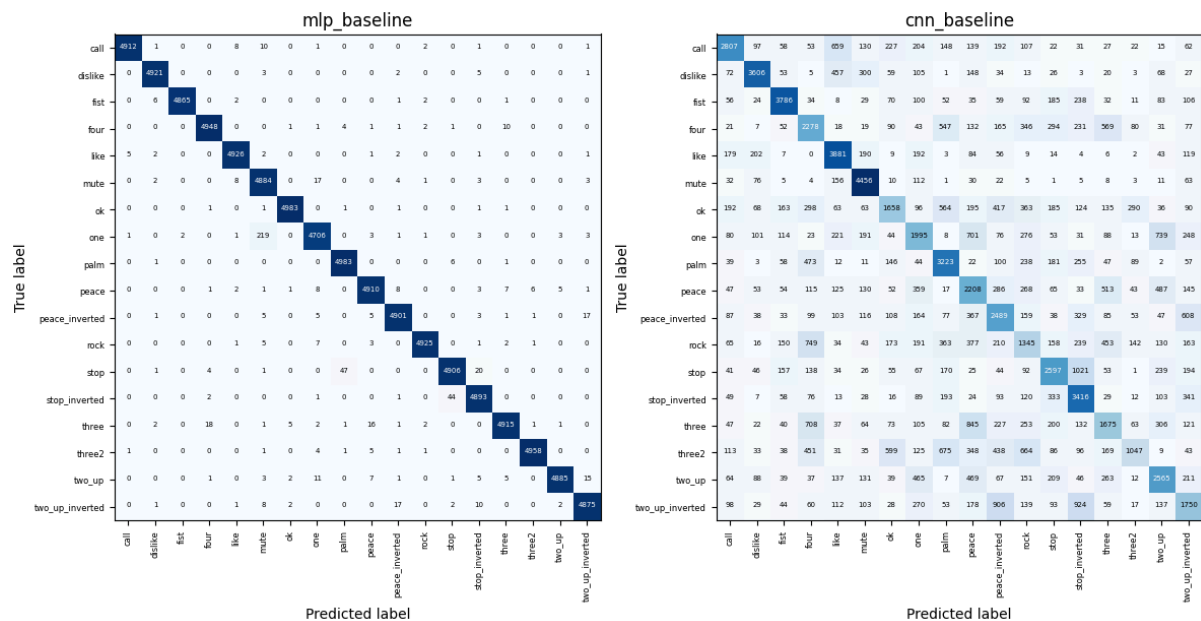
The MLP's near-perfect accuracy indicates that geometric pose cues alone suffice for almost all gestures in HaGRID.

(2) The CNN baseline lacks capacity.

Compared to the HaGRID baselines (e.g., ResNet-34 at ~91% F1), the project's CNN achieves substantially lower performance, confirming the model is too small to extract semantic features from RGB images. The HaGRID paper strongly supports this: page 6 lists models with tens of millions of parameters, whereas the project's CNN has only a small fraction of that.

Overall, the preliminary results validate the baseline pipeline while clearly motivating more powerful architectures for the RGB modality.

3.5 Confusion Matrix Analysis (Brief)



The confusion matrices above provide additional insight into class-level behavior.

(1) MLP (landmarks).

The matrix is almost perfectly diagonal, confirming the near-perfect accuracy. Misclassifications are extremely rare and occur mostly between gestures with very similar landmark structures (e.g., "three" vs. "three2," or "peace" vs. "one"). Overall, the landmarks encode gesture geometry so well that the MLP cleanly separates all 18 classes.

(2) CNN (RGB).

The CNN matrix shows much heavier confusion. The model frequently mixes pairs of gestures that differ mainly by orientation or subtle finger positions (e.g., "peace" ↔ "peace_inverted," "stop" ↔ "stop_inverted," "two_up" ↔ "two_up_inverted"). It also tends to misclassify detailed gestures such as "rock," "call," or "one," reflecting the limited capacity of the shallow CNN and the low-resolution inputs. These patterns are consistent with the overall low accuracy and highlight the need for a deeper pretrained CNN in the next stage.

4 Next Steps

4.1 Improving the Landmark Model

Implement a **Graph Neural Network (GNN)** for landmark processing.

A GNN can explicitly model:

- finger-bone connectivity,
- spatial relations between joints,
- invariances to rotation and deformation.

Since landmarks naturally form a hand graph, a GNN may improve performance on gestures where subtle geometric differences matter (e.g., “peace” vs. “peace inverted”).

4.2 Improving the RGB Model

Replace the simple CNN with stronger architectures, likely pretrained on ImageNet or trained on HaGRID itself:

- ResNet-18/34
- MobileNetV3
- EfficientNet
- ViT variants

The HaGRID benchmark (Table 2 on page 6) shows these models achieve 80–91% F1, establishing realistic performance targets.

Further improvements will include:

- Data augmentation (brightness, flipping, random cropping), consistent with the lighting-diversity experiments shown in Figures 5–6 (pages 7–8).
- Higher input resolution (e.g., 128×128 or 224×224).

4.3 Multimodal Fusion

After training improved models for each modality, the next step is **combining RGB features and landmark features** through late fusion of logits.

This multimodal approach may outperform either modality alone, especially for ambiguous gestures or those influenced by orientation or appearance.

4.4 Expanded Evaluation

Planned evaluations include:

- confusion matrices for deeper error visualization,
- per-gesture difficulty analysis (similar to HaGRID’s class-separability plots on page 5),

